UP LT GRADE · COMPUTER SCIENCE MAINS

Computer Graphics — High-Probability Numerical Questions with Full Solutions

Selected for Likelihood of Appearing, Based on the Confirmed Exam Pattern

Coverage: Line & circle rasterization, 2D transformations, windowing/clipping, frame buffer sizing, and curve evaluation.

Each question is tagged with its appearance likelihood, and every solution shows the full step-by-step method — exactly how it should be reproduced under exam word/time limits.

Practice set · step-by-step solutions · exam-style presentation

Table of Contents

TYPE 1 • LINE RASTERIZATION

Q1. DDA Line Drawing Algorithm

High Chance

Question

Using the DDA (Digital Differential Analyzer) algorithm, find all pixel positions for a line from (1,1) to (6,4).

Step 1: Compute Δx, Δy and Steps

$\Delta x = 6-1 = 5$, $\Delta y = 4-1 = 3$

Since $|\Delta x| > |\Delta y|$, Steps = $|\Delta x|$ = **5**

x-increment = Δx/Steps = 5/5 = 1   y-increment = Δy/Steps = 3/5 = **0.6**

Step 2: Generate Points (Round y at Each Step)

Step	x	y (exact)	y (rounded)
0	1	1.0	1
1	2	1.6	2
2	3	2.2	2
3	4	2.8	3
4	5	3.4	3
5	6	4.0	4

Final Answer

Plotted Pixels: **(1,1), (2,2), (3,2), (4,3), (5,3), (6,4)**

Why This Is High-Probability

DDA is almost always asked alongside Bresenham's so examiners can test whether you understand WHY Bresenham's (integer-only) is preferred over DDA's floating-point rounding.

TYPE 2 • LINE RASTERIZATION

Q2. Bresenham's Line Algorithm

Very High Chance

Question

Using Bresenham's Line Algorithm, find all pixel positions for a line from (2,2) to (9,6).

Step 1: Compute Initial Decision Parameter

$\Delta x = 9-2 = 7, \quad \Delta y = 6-2 = 4$

$p_0 = 2\Delta y - \Delta x = 2(4) - 7 = \mathbf{1}$

Step 2: Tabulate Each Decision

k	p_k	Decision	Next Pixel
0	1	$p \geq 0$: $+2\Delta y - 2\Delta x$	(3,3)
1	−5	$p < 0$: $+2\Delta y$	(4,3)
2	3	$p \geq 0$: $+2\Delta y - 2\Delta x$	(5,4)
3	−3	$p < 0$: $+2\Delta y$	(6,4)
4	5	$p \geq 0$: $+2\Delta y - 2\Delta x$	(7,5)
5	−1	$p < 0$: $+2\Delta y$	(8,5)
6	7	$p \geq 0$	(9,6)

Final Answer

Plotted Pixels: **(2,2), (3,3), (4,3), (5,4), (6,4), (7,5), (8,5), (9,6)**

Why This Is High-Probability

This is THE single most repeated numerical in Computer Graphics papers — always present the (k, p_k, Decision, Next Pixel) table exactly as shown, since the tabulated working carries most of the marks, not just the final pixel list.

TYPE 3 · CIRCLE RASTERIZATION

Q3. Midpoint Circle Algorithm

High Chance

Question

Using the Midpoint Circle Algorithm, compute the pixel positions for ONE octant of a circle with radius r = 8, centered at the origin.

Step 1: Initial Values

$p_0 = 1 - r = 1 - 8 = \mathbf{-7}$ Start: $(x_0,y_0) = (0,8)$

Step 2: Tabulate Each Decision (Stop When x ≥ y)

k	p_k	Decision	Next Point
0	−7	p<0: p+2x+1	(1,8)
1	−4	p<0: p+2x+1	(2,8)
2	1	p≥0: p+2x+1−2y	(3,7)
3	−6	p<0: p+2x+1	(4,7)
4	3	p≥0: p+2x+1−2y	(5,6)
5	2	p≥0: x≥y reached, stop	(6,5)

Final Answer

Octant Points: **(0,8), (1,8), (2,8), (3,7), (4,7), (5,6), (6,5)**

The remaining 7 octants (56 total points on the circle) are obtained instantly using 8-way symmetry — reflecting each (x,y) to (±x,±y) and (±y,±x).

Why This Is High-Probability

Always explicitly invoke "8-way symmetry" in your answer — examiners specifically check whether you understand that only 1/8th of the circle needs computing, not just whether you can execute the recurrence.

TYPE 4 · 2D TRANSFORMATIONS

Q4. 2D Translation

Very High Chance

Question

A point P(5, 8) is translated by $t_x = 3$, $t_y = -4$. Find the new coordinates P'.

Apply the Translation Formula

Formula

$x' = x + t_x \quad y' = y + t_y$

$x' = 5 + 3 = \mathbf{8} \quad y' = 8 + (-4) = \mathbf{4}$

Final Answer

P' = **(8, 4)**

TYPE 5 • 2D TRANSFORMATIONS

Q5. 2D Rotation About the Origin

Very High Chance

Question

A point P(4, 0) is rotated by $\theta = 90°$ about the origin. Find the new coordinates P'.

Apply the Rotation Formula

Formula

$x' = x \cdot \cos\theta - y \cdot \sin\theta \quad y' = x \cdot \sin\theta + y \cdot \cos\theta$

At $\theta=90°$: $\cos 90°=0$, $\sin 90°=1$

$x' = 4(0) - 0(1) = \mathbf{0}$

$y' = 4(1) + 0(0) = \mathbf{4}$

Final Answer

P' = **(0, 4)** — the point rotated 90° counter-clockwise, as expected (the positive X-axis point maps onto the positive Y-axis).

TYPE 6 · 2D TRANSFORMATIONS

Q6. 2D Scaling

Very High Chance

Question

A point P(3, 5) is scaled with factors $s_x = 2$, $s_y = 3$ relative to the origin. Find the new coordinates P'.

Apply the Scaling Formula

Formula

$x' = x \cdot s_x \quad y' = y \cdot s_y$

$x' = 3 \times 2 = \mathbf{6} \quad y' = 5 \times 3 = \mathbf{15}$

Final Answer

P' = **(6, 15)**

TYPE 7 · COMPOSITE TRANSFORMATIONS

Q7. Composite Transformation (Rotate then Translate)

High Chance

Question

A point P(1, 0) is first rotated by 90° about the origin, and then translated by (2, 3). Find the final coordinates. Explain why the ORDER of transformations matters.

Step 1: Apply Rotation First (90°)

$x' = 1(\cos 90°) - 0(\sin 90°) = 1(0) - 0(1) = \mathbf{0}$

$y' = 1(\sin 90°) + 0(\cos 90°) = 1(1) + 0(0) = \mathbf{1}$

After rotation: (0, 1)

Step 2: Apply Translation (2, 3)

$x'' = 0 + 2 = \mathbf{2}$ $y'' = 1 + 3 = \mathbf{4}$

Final Answer

Final Point = **(2, 4)**

Order matters: if we had TRANSLATED first (1,0)→(3,3), then ROTATED 90° about the origin, we'd get $x'=3(0)-3(1)=-3$, $y'=3(1)+3(0)=3$, giving (−3,3) — a COMPLETELY different result. Matrix multiplication is not commutative, so composite transformation order must always match the matrix multiplication order T·R·P (applied right to left).

TYPE 8 · WINDOWING

Q8. Window-to-Viewport Mapping

High Chance

Question

A window spans (0,0) to (20,20). It is mapped to a viewport spanning (0,0) to (100,100). Find the viewport coordinates of the window point (5, 10).

Apply the Window-to-Viewport Formula

Formula

$x_v = x_{vmin} + (x_w - x_{wmin}) \times$ (Viewport Width / Window Width)

$y_v = y_{vmin} + (y_w - y_{wmin}) \times$ (Viewport Height / Window Height)

Scale factor = 100/20 = 5 (both X and Y, since both window and viewport are square here)

$x_v = 0 + (5-0) \times 5 = \mathbf{25}$

$y_v = 0 + (10-0) \times 5 = \mathbf{50}$

Final Answer

Viewport Coordinates = **(25, 50)**

TYPE 9 · CLIPPING

Q9. Cohen-Sutherland Region Codes

High Chance

Question

A clipping window has x_{min}=10, x_{max}=50, y_{min}=10, y_{max}=50. Assign 4-bit region codes (order: Top, Bottom, Right, Left) to points P1(5, 60) and P2(30, 30), and determine whether the line P1P2 can be trivially accepted or rejected.

Step 1: Assign Region Code to P1(5, 60)

$x=5 < x_{min}=10 \rightarrow$ Left bit = 1

$y=60 > y_{max}=50 \rightarrow$ Top bit = 1

Code for P1 = **1001**

Step 2: Assign Region Code to P2(30, 30)

x=30 is between 10 and 50 → no Left/Right bit

y=30 is between 10 and 50 → no Top/Bottom bit

Code for P2 = **0000** (inside the window)

Step 3: Apply Trivial Accept/Reject Tests

Trivial ACCEPT test: both codes = 0000? NO (P1 is 1001)

Trivial REJECT test: bitwise AND of both codes ≠ 0000? 1001 AND 0000 = 0000 → NOT rejected

Final Answer

Neither trivial accept nor trivial reject applies — the line P1P2 is **partially inside** the window and must be clipped against the window boundaries (starting with the Top edge, since P1's Top bit is set).

TYPE 10 · DISPLAY SYSTEMS

Q10. Frame Buffer Size Calculation

Moderate Chance

Question

A display has a resolution of 800 × 600 pixels, using 16 bits per pixel. Calculate the required Frame Buffer size in Megabytes (MB).

Step 1: Total Number of Pixels

Total Pixels = 800 × 600 = **480,000 pixels**

Step 2: Total Bits and Conversion

Formula

Frame Buffer Size = Total Pixels × Bits per Pixel

Total bits = 480,000 × 16 = 7,680,000 bits
Total bytes = 7,680,000 / 8 = 960,000 bytes
In KB = 960,000 / 1024 = 937.5 KB
In MB = 937.5 / 1024 = **≈ 0.915 MB**

Final Answer

Required Frame Buffer Size ≈ **0.915 MB** (937.5 KB)

TYPE 11 · CURVES

Q11. Bezier Curve Point Evaluation

Moderate Chance

Question

A cubic Bezier curve has control points P0(0,0), P1(1,3), P2(3,3), P3(4,0). Find the point on the curve at $t = 0.5$.

Step 1: Bezier Curve Formula

Formula

$B(t) = (1-t)^3 P0 + 3(1-t)^2 t \cdot P1 + 3(1-t)t^2 \cdot P2 + t^3 P3$

Step 2: Compute Each Blending Coefficient at t=0.5

$(1-t)^3 = 0.5^3 = 0.125$

$3(1-t)^2 t = 3 \times 0.25 \times 0.5 = 0.375$

$3(1-t)t^2 = 3 \times 0.5 \times 0.25 = 0.375$

$t^3 = 0.5^3 = 0.125$

(Check: 0.125+0.375+0.375+0.125 = 1.0 ✓)

Step 3: Apply Coefficients to X and Y Separately

$X = 0.125(0) + 0.375(1) + 0.375(3) + 0.125(4) = 0 + 0.375 + 1.125 + 0.5 = \mathbf{2.0}$

$Y = 0.125(0) + 0.375(3) + 0.375(3) + 0.125(0) = 1.125 + 1.125 = \mathbf{2.25}$

Final Answer

Point on curve at t=0.5 = **(2.0, 2.25)**

— End of Practice Set —